

SUPPLEMENTARY MATERIALS: Supplementary Materials for: The AHI family of sum of squares polynomials

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Appendix A. Explicit SOS decompositions of AHI polynomials.

Table A.1: SOS decompositions of AHI polynomials. The last two rows are the factors f_2 and f_1 used in the worked example of Section 5.6.

AHI polynomial	SOS representation
$2(x_1^4 + x_2^4) - 4x_1^2x_2^2$	$2(x_1^2 - x_2^2)^2$
$(x_1^4 + x_2^4)x_3^2 + 2(x_1^4 + x_3^4)x_2^2$ $+ 2(x_2^4 + x_3^4)x_1^2 - 10x_1^2x_2^2x_3^2$	$(x_1^2 - x_2^2)^2x_3^2 + 2(x_1^2 - x_3^2)^2x_2^2$ $+ 2(x_2^2 - x_3^2)^2x_1^2$
$(x_1^4x_2^4 + x_2^4x_3^4)x_3^2$ $+ 2(x_1^4x_2^4 + x_3^4)x_2^2x_3^2$ $+ 0.5(x_2^4x_3^4 + x_3^4)x_1^2x_2^2$ $- 3.75x_1^2x_2^4x_3^4$	$(x_1^2x_2^2 - x_2^2x_3^2)^2x_3^2$ $+ 2(x_1^2x_2^2 - x_3^2)^2x_2^2x_3^2$ $+ 0.5(x_2^2x_3^2 - x_3^2)^2x_1^2x_2^2$ $+ 3.25x_1^2x_2^4x_3^4$
$6(x_1^4x_3^4 + x_2^8x_4^4) - 2x_1^2x_2^4x_3^2x_4^2$	$6(x_1^2x_3^2 - x_2^4x_4^2)^2 + 10x_1^2x_2^4x_3^2x_4^2$
$x_3^2x_4^2(x_1^4 + x_2^4) + x_2^2x_4^2(x_1^4 + x_3^4)$ $+ x_2^2x_3^2(x_1^4 + x_4^4) + x_1^2x_4^2(x_2^4 + x_3^4)$ $+ x_1^2x_3^2(x_2^4 + x_4^4) + x_1^2x_2^2(x_3^4 + x_4^4)$ $- 3x_1^2x_2^2x_3^2x_4^2$	$x_3^2x_4^2(x_1^2 - x_2^2)^2 + x_2^2x_4^2(x_1^2 - x_3^2)^2 +$ $x_2^2x_3^2(x_1^2 - x_4^2)^2 + x_1^2x_4^2(x_2^2 - x_3^2)^2 +$ $x_1^2x_3^2(x_2^2 - x_4^2)^2 + x_1^2x_2^2(x_3^2 - x_4^2)^2$ $+ 9x_1^2x_2^2x_3^2x_4^2$

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Table A.1 – continued from previous page

AHI polynomial	SOS representation
$2x_2^2x_3^2x_4^2(x_1^4 + x_5^4)$ $+2x_1^2x_3^2x_4^2(x_2^4 + x_5^4)$ $+2x_1^2x_2^2x_4^2(x_3^4 + x_5^4)$ $+2x_1^2x_2^2x_3^2(x_4^4 + x_5^4)$ $+x_3^2x_4^2x_5^2(x_1^4 + x_2^4)$ $+x_2^2x_4^2x_5^2(x_1^4 + x_3^4)$ $+x_2^2x_3^2x_5^2(x_1^4 + x_4^4)$ $+x_1^2x_4^2x_5^2(x_2^4 + x_3^4)$ $+x_1^2x_3^2x_5^2(x_2^4 + x_4^4)$ $+x_1^2x_2^2x_5^2(x_3^4 + x_4^4)$ $-4x_1^2x_2^2x_3^2x_4^2x_5^2$	$2x_2^2x_3^2x_4^2(x_1^2 - x_5^2)^2$ $+2x_1^2x_3^2x_4^2(x_2^2 - x_5^2)^2 +$ $2x_1^2x_2^2x_4^2(x_3^2 - x_5^2)^2$ $+2x_1^2x_2^2x_3^2(x_4^2 - x_5^2)^2 +$ $x_3^2x_4^2x_5^2(x_1^2 - x_2^2)^2$ $+x_2^2x_4^2x_5^2(x_1^2 - x_3^2)^2 +$ $x_2^2x_3^2x_5^2(x_1^2 - x_4^2)^2$ $+x_1^2x_4^2x_5^2(x_2^2 - x_3^2)^2 +$ $x_1^2x_3^2x_5^2(x_2^2 - x_4^2)^2$ $+x_1^2x_2^2x_5^2(x_3^2 - x_4^2)^2$ $+24x_1^2x_2^2x_3^2x_4^2x_5^2$

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Appendix B. Hessian computation for Theorem 3.1.

Consider the structure

$$(B.1) \quad f(x) = \lambda_1 \lambda_2 (x_1^{\alpha_1} x_2^{\beta_1} + x_1^{\alpha_2} x_2^{\beta_2}) + (\lambda_1^2 + \lambda_2^2 - c) x_1^{\alpha_1 + \alpha_2} x_2^{\beta_1 + \beta_2},$$

where $\lambda_1, \lambda_2 \geq 0$ and $\alpha_i, \beta_i \geq 2$ are even. If f is convex then all principal minors of the Hessian $H(x)$ are non-negative.

Consider H_{11} . If $\lambda_1^2 + \lambda_2^2 - c$ is positive then all terms are positive and H_{11} is SOS. If $\lambda_1^2 + \lambda_2^2 - c$ is negative, then H_{11} is non-negative if and only if it is SOS, which requires

$$\begin{aligned} & (\lambda_1^2 + \lambda_2^2 - c)(\alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 - 1) \\ = & 2\sqrt{\lambda_1 \lambda_2 (2\alpha_1)(2\alpha_1 - 1)} \sqrt{\lambda_1 \lambda_2 (2\alpha_2)(2\alpha_2 - 1)} \\ = & 2\lambda_1 \lambda_2 \sqrt{(2\alpha_1)(2\alpha_1 - 1)(2\alpha_2)(2\alpha_2 - 1)}, \end{aligned}$$

a contradiction, since the left-hand side is negative while the right-hand side is positive. The same holds for H_{22} . Hence the first necessary condition for convexity of f is

$$\lambda_1^2 + \lambda_2^2 - c > 0.$$

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We now compute $H_{11}H_{22} - H_{12}^2$:

$$\begin{aligned}
& H_{11}H_{22} - H_{12}^2 \\
&= \lambda_1^2 \lambda_2^2 (4\alpha_1 \beta_1) (2\alpha_1 - 1) (2\beta_1 - 1) x_1^{4\alpha_1-2} x_2^{4\beta_1-2} \\
&\quad + \lambda_1^2 \lambda_2^2 (4\alpha_2 \beta_1) (2\alpha_2 - 1) (2\beta_1 - 1) \\
&\quad \times x_1^{2\alpha_1+2\alpha_2-2} x_2^{2\beta_1+2\beta_2-2} \\
&\quad + \lambda_1 \lambda_2 (\lambda_1^2 + \lambda_2^2 - c) (\alpha_1 + \alpha_2) (\alpha_1 + \alpha_2 - 1) \\
&\quad \times (2\beta_1) (2\beta_1 - 1) x_1^{3\alpha_1+\alpha_2-2} x_2^{3\beta_1+\beta_2-2} \\
&\quad + \lambda_1^2 \lambda_2^2 (4\alpha_1 \beta_2) (2\alpha_1 - 1) (2\beta_2 - 1) \\
&\quad \times x_1^{2\alpha_1+2\alpha_2-2} x_2^{2\beta_1+2\beta_2-2} \\
&\quad + \lambda_1^2 \lambda_2^2 (4\alpha_2 \beta_2) (2\alpha_2 - 1) (2\beta_2 - 1) x_1^{4\alpha_2-2} x_2^{4\beta_2-2} \\
&\quad + \lambda_1 \lambda_2 (\lambda_1^2 + \lambda_2^2 - c) (\alpha_1 + \alpha_2) (\alpha_1 + \alpha_2 - 1) \\
&\quad \times (2\beta_2) (2\beta_2 - 1) x_1^{\alpha_1+3\alpha_2-2} x_2^{\beta_1+3\beta_2-2} \\
&\quad + \lambda_1 \lambda_2 (\lambda_1^2 + \lambda_2^2 - c) (2\alpha_1) (2\alpha_1 - 1) \\
&\quad \times (\beta_1 + \beta_2) (\beta_1 + \beta_2 - 1) x_1^{3\alpha_1+\alpha_2-2} x_2^{3\beta_1+\beta_2-2} \\
&\quad + \lambda_1 \lambda_2 (\lambda_1^2 + \lambda_2^2 - c) (2\alpha_2) (2\alpha_2 - 1) \\
&\quad \times (\beta_1 + \beta_2) (\beta_1 + \beta_2 - 1) x_1^{\alpha_1+3\alpha_2-2} x_2^{\beta_1+3\beta_2-2} \\
&\quad + (\lambda_1^2 + \lambda_2^2 - c)^2 (\alpha_1 + \alpha_2) (\alpha_1 + \alpha_2 - 1) \\
&\quad \times (\beta_1 + \beta_2) (\beta_1 + \beta_2 - 1) x_1^{2\alpha_1+2\alpha_2-2} x_2^{2\beta_1+2\beta_2-2} \\
&\quad - \lambda_1^2 \lambda_2^2 (16\alpha_1^2 \beta_1^2) x_1^{4\alpha_1-2} x_2^{4\beta_1-2} \\
&\quad - \lambda_1^2 \lambda_2^2 (16\alpha_2^2 \beta_2^2) x_1^{4\alpha_2-2} x_2^{4\beta_2-2} \\
&\quad - 2\lambda_1^2 \lambda_2^2 (4\alpha_1 \beta_1) (4\alpha_2 \beta_2) x_1^{2\alpha_1+2\alpha_2-2} x_2^{2\beta_1+2\beta_2-2} \\
&\quad - 2\lambda_1 \lambda_2 (\lambda_1^2 + \lambda_2^2 - c) (4\alpha_1 \beta_1) \\
&\quad \times (\alpha_1 + \alpha_2) (\beta_1 + \beta_2) x_1^{3\alpha_1+\alpha_2-2} x_2^{3\beta_1+\beta_2-2} \\
&\quad - 2\lambda_1 \lambda_2 (\lambda_1^2 + \lambda_2^2 - c) (4\alpha_2 \beta_2) \\
&\quad \times (\alpha_1 + \alpha_2) (\beta_1 + \beta_2) x_1^{\alpha_1+3\alpha_2-2} x_2^{\beta_1+3\beta_2-2} \\
&\quad - (\lambda_1^2 + \lambda_2^2 - c)^2 (\alpha_1 + \alpha_2)^2 \\
&\quad \times (\beta_1 + \beta_2)^2 x_1^{2\alpha_1+2\alpha_2-2} x_2^{2\beta_1+2\beta_2-2}.
\end{aligned}$$

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19 We now collect the coefficients of equal monomials.

20 Coefficient of $x_1^{4\alpha_1-2} x_2^{4\beta_1-2}$:

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$$\begin{aligned}
& \lambda_1^2 \lambda_2^2 (4\alpha_1 \beta_1) (2\alpha_1 - 1) (2\beta_1 - 1) - 16\lambda_1^2 \lambda_2^2 \alpha_1^2 \beta_1^2 \\
&= \lambda_1^2 \lambda_2^2 (4\alpha_1 \beta_1) (-2\alpha_1 - 2\beta_1 + 1) < 0,
\end{aligned}$$

22 since $\alpha_1, \beta_1 \geq 2$.

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23 Coefficient of $x_1^{4\alpha_2-2}x_2^{4\beta_2-2}$:

$$\begin{aligned} & \lambda_1^2\lambda_2^2(4\alpha_2\beta_2)(2\alpha_2-1)(2\beta_2-1) - 16\lambda_1^2\lambda_2^2\alpha_2^2\beta_2^2 \\ & = \lambda_1^2\lambda_2^2(4\alpha_2\beta_2)(-2\alpha_2-2\beta_2+1) < 0, \end{aligned}$$

25 since $\alpha_2, \beta_2 \geq 2$.

26 Coefficient of $x_1^{3\alpha_1+\alpha_2-2}x_2^{3\beta_1+\beta_2-2}$: writing $s^* = \lambda_1\lambda_2(\lambda_1^2 + \lambda_2^2 - c)$,

$$\begin{aligned} & s^*(\alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 - 1)(2\beta_1)(2\beta_1 - 1) \\ & + s^*(2\alpha_1)(2\alpha_1 - 1)(\beta_1 + \beta_2)(\beta_1 + \beta_2 - 1) \\ & - 2s^*(2\alpha_1)(2\beta_1)(\alpha_1 + \alpha_2)(\beta_1 + \beta_2) \\ & = s^*(2\alpha_2\beta_1 - 2\alpha_1\beta_2)^2 \\ & - s^*(\alpha_1 + \alpha_2)(2\beta_1)(\alpha_1 + \alpha_2 + 2\beta_1 - 1) \\ & - s^*(2\alpha_1)(\beta_1 + \beta_2)(2\alpha_1 + \beta_1 + \beta_2 - 1) < 0. \end{aligned}$$

28 The coefficient of $x_1^{\alpha_1+3\alpha_2-2}x_2^{\beta_1+3\beta_2-2}$ is negative by the same computation.

29 Coefficient of $x_1^{2\alpha_1+2\alpha_2-2}x_2^{2\beta_1+2\beta_2-2}$:

$$\begin{aligned} & \lambda_1^2\lambda_2^2(2\alpha_2)(2\beta_1)(2\alpha_2-1)(2\beta_1-1) \\ & + \lambda_1^2\lambda_2^2(2\alpha_1)(2\alpha_1-1)(2\beta_2)(2\beta_2-1) \\ & - 2\lambda_1^2\lambda_2^2(2\alpha_1)(2\beta_1)(2\alpha_2)(2\beta_2) \\ & + (\lambda_1^2 + \lambda_2^2 - c)^2(\alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 - 1) \\ & \times (\beta_1 + \beta_2)(\beta_1 + \beta_2 - 1) \\ & - (\lambda_1^2 + \lambda_2^2 - c)^2(\alpha_1 + \alpha_2)^2(\beta_1 + \beta_2)^2. \end{aligned}$$

31 Splitting this into two parts, the first,

$$\begin{aligned} & \lambda_1^2\lambda_2^2(2\alpha_2)(2\beta_1)(2\alpha_2-1)(2\beta_1-1) \\ & + \lambda_1^2\lambda_2^2(2\alpha_1)(2\alpha_1-1)(2\beta_2)(2\beta_2-1) \\ & - 2\lambda_1^2\lambda_2^2(2\alpha_1)(2\beta_1)(2\alpha_2)(2\beta_2) < 0, \end{aligned}$$

33 is negative by the computation used for $x_1^{3\alpha_1+\alpha_2-2}x_2^{3\beta_1+\beta_2-2}$, and the second,

$$\begin{aligned} & (\lambda_1^2 + \lambda_2^2 - c)^2 \left[(\alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 - 1) \right. \\ & \times (\beta_1 + \beta_2)(\beta_1 + \beta_2 - 1) \\ & \left. - (\alpha_1 + \alpha_2)^2(\beta_1 + \beta_2)^2 \right] < 0, \end{aligned}$$

35 is negative by the computation used for $x_1^{4\alpha_1-2}x_2^{4\beta_1-2}$.

36 Every collected coefficient is therefore negative, proving

$$37 \quad H_{11}H_{22} - H_{12}^2 < 0$$

38 and hence that polynomials of the form (B.1) are functionally non-convex.

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